Chapter 4 — Stoichiometry

Exercise Solutions · Full textbook exercise · PECTAA Chemistry 11 · Book pp. 88–90

Q.1 — Multiple Choice Questions (with correct option and reason)

I. Which one of the following statements is incorrect?

(a) One mole of nitrogen gas contains Avogadro's number of molecules

(b) One mole of ozone gas contains Avogadro's number of molecules

(c) One mole of ozone contains Avogadro's number of O atoms

(d) One mole of hydrogen gas contains Avogadro's number of molecules

Answer: (c) One mole of ozone contains Avogadro's number of O atoms — One mole of O_3 contains N_a molecules but $3N_a$ oxygen atoms, so statement (c) is incorrect.

II. Which one of the following has greatest mass?

(a) 0.5 mol of N_2

(b) 0.5 mol of NH_3

(c) 0.5 mol of He

(d) 0.5 mol of CO_2

Answer: (d) 0.5 mol of CO_2 — Mass = n × M: N_2 14 g, NH_3 8.5 g, He 2 g, CO_2 22 g. CO_2 is greatest.

III. Which one of the following gases will have greatest volume at STP?

(a) 22 g of CO_2

(b) 88 g of N_2O

(c) 28 g of CO

(d) 28 g of N_2

Answer: (b) 88 g of N_2O — Moles: CO_2 0.5, N_2O 2.0, CO 1.0, N_2 1.0. Volume ∝ moles, so 88 g N_2O (2 mol) is greatest.

IV. Which of the following contains same number of particles as present in 12 g of carbon?

(a) 28 g of iron (Ar Fe = 56)

(b) 48 g of magnesium (Ar Mg = 24)

(c) 32 g of S_8 molecules (Ar S = 32)

(d) 40 g of carbon dioxide (M CO_2 = 44)

Answer: (c) 32 g of S_8 molecules (Ar S = 32) — 12 g C = 1 mol C atoms. 32 g S_8 = 0.125 mol S_8 molecules = 1 mol S atoms, i.e. the same number of atoms.

The question says "particles"; the match is on number of atoms. If "particles" is taken to mean S_8 molecules, none of the options matches exactly.

V. Volume at S.T.P. of 22 g of CO_2 is same as that of:

(a) 2 g of hydrogen

(b) 8.5 g of NH_3

(c) 64 g of gaseous SO_2

(d) 7 g of CO

Answer: (b) 8.5 g of NH_3 — 22 g CO_2 = 0.5 mol → 11.2 dm^3. 8.5 g NH_3 = 0.5 mol → the same volume.

VI. 4.0 g of NaOH (M = 40) contains same number of sodium ions as are present in:

(a) 10.6 g of Na_2CO_3 (M = 106)

(b) 58.5 g of NaCl (M = 58.5)

(c) 76 g Na_2SO_4 (M = 142)

(d) 8.5 g of $NaNO_3$ (M = 85)

studysapiens.com

Answer: (d) 8.5 g of $NaNO_3$ (M = 85) — 4.0 g NaOH = 0.10 mol Na^+. 8.5 g $NaNO_3$ = 0.10 mol Na^+. Na_2CO_3 gives 0.20 mol Na^+.

VII. A container holds 0.5 moles of an ideal gas at STP. What is the volume of the gas in dm^3?

(a) 11.2 dm^3 **(b)** 22.4 dm^3
(c) 44.8 dm^3 **(d)** 12.2 dm^3

Answer: (a) 11.2 dm^3 — V = 0.5 × 22.4 = 11.2 dm^3.

VIII. A solution contains 4.0 g NaOH (M = 40.0) in 250 cm^3 of solution. What is the molar concentration?

(a) 0.10 mol dm^{-3} **(b)** 0.20 mol dm^{-3}
(c) 0.40 mol dm^{-3} **(d)** 0.80 mol dm^{-3}

Answer: (c) 0.40 mol dm^{-3} — n = 4.0/40 = 0.10 mol; V = 0.250 dm^3; c = 0.10/0.250 = 0.40 mol dm^{-3}.

IX. A solution contains 10.0 g of an unknown solute in 250 cm^3. If the molar concentration is 0.20 mol dm^{-3}, what is the molar mass?

(a) 50 g mol^{-1} **(b)** 100 g mol^{-1}
(c) 200 g mol^{-1} **(d)** 400 g mol^{-1}

Answer: (c) 200 g mol^{-1} — n = cV = 0.20 × 0.250 = 0.050 mol; M = 10.0/0.050 = 200 g mol^{-1}.

X. A sample of N_2 (M = 28.0) has a mass of 14.0 g. How many nitrogen atoms are present?

(a) 3.01×10^{23} atoms **(b)** 6.02×10^{23} atoms
(c) 1.50×10^{24} atoms **(d)** 2.40×10^{24} atoms

Answer: (b) 6.02×10^{23} atoms — 14.0 g N_2 = 0.50 mol molecules = 1.0 mol N atoms = 6.02×10^{23} atoms.

XI. A gas has a density of 1.43 g dm^{-3} at STP. What is the molar mass? (Molar volume = 22.4 dm^3 mol^{-1})

(a) 14.3 g mol^{-1} **(b)** 22.4 g mol^{-1}
(c) 32.0 g mol^{-1} **(d)** 64.0 g mol^{-1}

Answer: (c) 32.0 g mol^{-1} — M = density × molar volume = 1.43 × 22.4 ≈ 32.0 g mol^{-1}.

XII. A gas has a density of 1.96 g dm^{-3} at STP (0 °C, 1.00 atm). What is its molecular mass?

(a) 11.2 g mol^{-1} **(b)** 22.4 g mol^{-1}
(c) 44.0 g mol^{-1} **(d)** 88.0 g mol^{-1}

Answer: (c) 44.0 g mol^{-1} — M = 1.96 × 22.4 ≈ 44.0 g mol^{-1} (CO_2).

studysapiens.com

Q.2 — Short Answer Questions

a. How is the concept of mole derived from Avogadro's number?

One mole is the amount of substance that contains exactly Avogadro's number (6.022×10^{23}) of elementary entities (atoms, molecules, ions, etc.). It is the amount containing as many particles as there are atoms in exactly 12 g of carbon-12.

b. Define with one example: (a) Molar mass (b) Molar volume (c) Molar concentration.

(a) Molar mass: mass of one mole of a substance (g mol^{-1}); e.g. H_2O = 18 g mol^{-1}. (b) Molar volume: volume occupied by one mole of a gas at given T and P; at STP ≈ 22.4 dm^3 mol^{-1}. (c) Molar concentration: moles of solute per dm^3 of solution (mol dm^{-3}); e.g. 1 mol NaCl in 1 dm^3 solution = 1 M.

c. What do you mean by molar volume of a gas? How Avogadro's number is related with molar volume?

Molar volume is the volume occupied by one mole of a gas. At STP it is about 22.4 dm^3. One mole contains Avogadro's number of molecules, so 22.4 dm^3 of any ideal gas at STP contains 6.022×10^{23} molecules (Avogadro's law).

d. 39 g of potassium and 56 g of iron have equal number of atoms in them. Justify.

Moles of K = 39/39 = 1 mol; moles of Fe = 56/56 = 1 mol. Equal moles of atoms contain equal numbers of atoms (N_a each).

e. 4 g He, 17 g NH_3 and 64 g SO_2 occupy separately 22.414 dm^3 although their sizes and molecular masses differ. Explain.

Each sample is one mole (He 4 g, NH_3 17 g, SO_2 64 g). By Avogadro's law, equal moles of any ideal gas at the same T and P occupy the same volume, because gas volume depends mainly on the number of particles and the empty space between them, not on molecular size or mass.

f. Do you think that 1 mole of H_2 and 1 mole of NH_3 at 0 °C and 1 atm will have Avogadro's number of particles?

Yes. One mole of any substance contains Avogadro's number of particles. Thus 1 mol H_2 contains N_a H_2 molecules and 1 mol NH_3 contains N_a NH_3 molecules (though the numbers of atoms differ).

g. What is stoichiometry? Give the basic assumptions of stoichiometric calculations.

Stoichiometry is the quantitative relationship between reactants and products in a balanced chemical equation. Assumptions: the equation is correctly balanced; reactants are pure; the reaction goes to completion as written; the limiting reactant controls the yield; and, for gases, ideal behaviour and stated T/P are used.

studysapiens.com

g. What is a limiting reactant? How does it control the quantity of the product formed?

The limiting reactant is the reactant that is completely consumed first and therefore limits the amount of product. Once it is used up the reaction stops, so the maximum (theoretical) yield is calculated from the moles of the limiting reactant.

The book repeats the letter (g) for this and the previous short question; both are answered.

h. Differentiate theoretical and actual yields. How is the percentage yield calculated?

Theoretical yield is the maximum mass of product predicted from the balanced equation and the limiting reactant. Actual yield is the mass actually obtained in the experiment. Percentage yield = (actual yield / theoretical yield) × 100.

i. What are the factors mostly responsible for the low yield of products in chemical reactions?

Incomplete reaction/reversibility; side reactions forming by-products; loss of product during transfer, filtration or purification; impure reactants; and experimental/measurement errors.

Descriptive / Long Questions

Q.3 Differentiate limiting and non-limiting reactants. How is a limiting reactant determined from a balanced equation and given data?

The limiting reactant is completely consumed and determines the maximum product; the non-limiting (excess) reactant remains after the reaction.

To identify it: (1) balance the equation; (2) convert given masses to moles; (3) divide the moles of each reactant by its stoichiometric coefficient; (4) the reactant with the smallest ratio is limiting. Alternatively, calculate the product each reactant could form; the one giving less product is limiting.

Example: $2H_2 + O_2 \rightarrow 2H_2O$. With 4 mol H_2 and 1 mol O_2, $H_2/2 = 2$ and $O_2/1 = 1$; O_2 is limiting and controls the water formed.

Q.4 Differentiate actual and theoretical yields. Why is the theoretical yield always greater than actual yield?

Theoretical yield is the calculated maximum from stoichiometry; actual yield is the experimentally obtained mass. Theoretical yield is greater because real reactions may be incomplete or reversible, side reactions consume reactants, and product is lost during handling/purification. Hence percentage yield is usually below 100%.

Numerical Problems (fully worked)

Q.5 A solution of NaOH is prepared by dissolving 2.00 g in water to make 250 cm^3. (a) Molar mass of NaOH. (b) Moles used. (c) Concentration in mol dm^{-3}. (d) Concentration after dilution to 500 cm^3.

Given: mass = 2.00 g; V_1 = 250 cm^3 = 0.250 dm^3; V_2 = 500 cm^3 = 0.500 dm^3

studysapiens.com

Formula: $n = m/M$; $c = n/V$; $c_1V_1 = c_2V_2$

(a) $M(NaOH) = 23 + 16 + 1 = 40.0$ g mol^{-1}.

(b) $n = 2.00/40.0 = 0.0500$ mol.

(c) $c = 0.0500/0.250 = 0.200$ mol dm^{-3}.

(d) $c_2 = c_1V_1/V_2 = (0.200 \times 0.250)/0.500 = 0.100$ mol dm^{-3}.

Answer: (a) 40.0 g mol^{-1}; (b) 0.0500 mol; (c) 0.200 mol dm^{-3}; (d) 0.100 mol dm^{-3}

Q.6 $4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$. 34.0 g NH_3 reacts with 96 g O_2. (a) Limiting reactant. (b) Maximum mass of NO. (c) Mass of excess reactant remaining.

Given: $m(NH_3) = 34.0$ g; $m(O_2) = 96$ g; $M(NH_3) = 17$, $M(O_2) = 32$, $M(NO) = 30$ g mol^{-1}

Formula: $n = m/M$; compare n/coefficient

$n(NH_3) = 34.0/17 = 2.00$ mol; $n(O_2) = 96/32 = 3.00$ mol.

Divide by coefficients: NH_3 $2.00/4 = 0.50$; O_2 $3.00/5 = 0.60$. NH_3 is smaller → NH_3 is limiting.

Moles NO = moles NH_3 (4:4) = 2.00 mol → mass NO = $2.00 \times 30 = 60.0$ g.

O_2 used = $(5/4) \times 2.00 = 2.50$ mol. O_2 remaining = $3.00 - 2.50 = 0.50$ mol → $0.50 \times 32 = 16.0$ g.

Answer: (a) NH_3 is limiting; (b) 60.0 g NO; (c) 16.0 g O_2 remains

Q.7 $Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(s) + 3CO_2(g)$. 1.00 kg Fe_2O_3 with excess CO gives 650 g Fe. Find percentage yield. (Ar O = 16.0, Fe = 55.8)

Given: $m(Fe_2O_3) = 1000$ g; actual Fe = 650 g; $M(Fe_2O_3) = 2(55.8) + 3(16) = 159.6$ g mol^{-1}

Formula: % yield = (actual / theoretical) × 100

$n(Fe_2O_3) = 1000/159.6 = 6.266$ mol.

Theoretical $n(Fe) = 2 \times 6.266 = 12.53$ mol → mass = $12.53 \times 55.8 = 699.3$ g.

% yield = $(650/699.3) \times 100 = 92.9\% \approx 93.0\%$.

Answer: Percentage yield of iron ≈ 93.0%

Q.8 1.5 g C_2H_6 is burnt in excess O_2: $2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$. Find volume of CO_2 at STP, and (a) molecules of CO_2, (b) molecules of O_2 reacted, (c) number of C–H bonds broken.

Given: $m(C_2H_6) = 1.5$ g; $M(C_2H_6) = 30$ g mol^{-1}; molar volume at STP = 22.4 dm^3 mol^{-1}; $N_a = 6.022 \times 10^{23}$

Formula: $n = m/M$; volume = $n \times 22.4$; particles = $n \times N_a$

$n(C_2H_6) = 1.5/30 = 0.050$ mol.

$n(CO_2) = (4/2) \times 0.050 = 0.100$ mol → $V = 0.100 \times 22.4 = 2.24$ dm^3.

studysapiens.com

(a) molecules CO_2 = 0.100 × 6.022 × 10^{23} = 6.02 × 10^{22}.

(b) n(O_2) = (7/2) × 0.050 = 0.175 mol → molecules = 0.175 × 6.022 × 10^{23} = 1.05 × 10^{23}.

(c) Each C_2H_6 has 6 C–H bonds; bonds broken = 0.050 × 6 × 6.022 × 10^{23} = 1.81 × 10^{23}.

Answer: V(CO_2) = 2.24 dm^3; (a) 6.02 × 10^{22} molecules; (b) 1.05 × 10^{23} molecules; (c) 1.81 × 10^{23} C–H bonds

Textbook check (p. 90): part (a) says "solid CO_2", but at STP CO_2 is a gas; the calculation uses gaseous CO_2.

Quick Check Solutions (in-chapter practice boxes)

Quick Check 4.1 (book p. 72)

a. Molar mass of $KMnO_4$.

39 + 55 + 4(16) = 158 g mol^{-1}.

b. Moles in 0.23 g sodium.

n = 0.23/23 = 0.010 mol.

c. Mass of 1.5 moles of $Ca(OH)_2$.

M = 40 + 2(16+1) = 74 g mol^{-1}; mass = 1.5 × 74 = 111 g.

d. Moles in 24.5 g $KClO_3$.

M = 39 + 35.5 + 48 = 122.5 g mol^{-1}; n = 24.5/122.5 = 0.20 mol.

e. Molecules in 1.75 g H_2O_2.

M = 34 g mol^{-1}; n = 1.75/34 = 0.0515 mol; molecules = 0.0515 × 6.022×10^{23} = 3.10×10^{22}.

f. Atoms in 15 g of a gold ring.

n = 15/197 = 0.0761 mol; atoms = 0.0761 × 6.022×10^{23} = 4.58×10^{22}.

Quick Check 4.2 (book p. 74)

a. A copper wire contains 27.10×10^{25} atoms. Calculate moles.

n = (27.10×10^{25})/(6.022×10^{23}) = 450 mol.

b. Molecules in 1×10^{-6} g isopentyl acetate ($C_7H_{14}O_2$); atoms of C, H, O.

M = 130 g mol^{-1}; n = 1×10^{-6}/130 = 7.69×10^{-9} mol; molecules = 4.63×10^{15}. C atoms = 7× = 3.24×10^{16}; H atoms = 14× = 6.48×10^{16}; O atoms = 2× = 9.26×10^{15}.

Quick Check 4.3 (book p. 76)

Molar mass of a gas with density 1.34 g dm^{-3} at STP.

M = density × molar volume = 1.34 × 22.4 = 30.0 g mol^{-1}.

studysapiens.com

Quick Check 4.4 (book p. 77)

Molar concentration of 7.9 g $KMnO_4$ in 1 dm^3 (M = 158).

n = 7.9/158 = 0.050 mol; c = 0.050/1 = 0.050 mol dm^{-3}.

Quick Check 4.5 (book p. 79)

Moles of CO_2 produced from 2.25 moles of glucose (excess O_2).

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$. Moles $CO_2 = 6 \times 2.25 = 13.5$ mol.

Quick Check 4.6 (book p. 79)

How much Fe can be produced from 160 g Fe_2O_3?

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$. $n(Fe_2O_3) = 160/159.6 \approx 1.00$ mol → $n(Fe) = 2.00$ mol → mass = $2.00 \times 55.8 \approx 112$ g.

Quick Check 4.7 (book p. 80)

Volume of CO_2 at STP when 4.5 dm^3 methane is burnt (excess O_2).

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$. Volume ratio CH_4:CO_2 = 1:1, so $V(CO_2) = 4.5$ dm^3.

Quick Check 4.8 (book p. 80)

Mass of NaOCl produced from 2.25 moles Cl_2 with excess NaOH.

$2NaOH + Cl_2 \rightarrow NaOCl + NaCl + H_2O$. $n(NaOCl) = 2.25$ mol; M = 74.5 g mol^{-1}; mass = $2.25 \times 74.5 = 167.6$ g.

Quick Check 4.9 (book p. 85)

Which mixture produces the greatest amount of NH_3? $N_2 + 3H_2 \rightarrow 2NH_3$.

(e) Each produces the same amount (2 mol NH_3). In (a) the ratio is stoichiometric; in (b), (c), (d) one reactant is in excess but the limiting reactant still gives 2 mol NH_3.

Quick Check 4.10 (book p. 87)

Percentage yield of CaO when 4.5 kg limestone gives 2.5 kg CaO.

$CaCO_3 \rightarrow CaO + CO_2$. Theoretical CaO = $4.5 \times (56/100) = 2.52$ kg. % yield = $(2.5/2.52) \times 100 = 99.2\%$.

Textbook checks flagged in this chapter

- p. 89, MCQ IV: the match with 12 g carbon is on number of atoms (32 g S_8 contains 1 mol S atoms); the wording “particles” is ambiguous.
- p. 90, Q.2: the letter (g) is printed twice; both short questions are answered.
- p. 90, Q.8(a): CO_2 is gaseous at STP, not solid.

studysapiens.com